

RVE Model with Porosity for 2D Woven CVI SiC_f/SiC Composites

Xiuli Shen and Longdong Gong

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A representative volume element (RVE) model with porosity for 2D woven chemical vapor infiltration (CVI) SiC_f/SiC composites is presented, and its mechanical properties are analyzed. Samples are divided after a tensile test, and their cross sections are scanned with a scanning electron microscope. The size of the feature structure of the RVE model is determined based on the measurement statistics of the feature structure parameters. In accordance with CVI technology, the deposition rates of the matrix in each direction along the surface of fiber bundles are assumed to be similar. The porosity structure is formed naturally when the RVE model is established. The RVE model conforms to the real structure and accurately shows the location and geometric shape of internal porosity. The relative error of the tensile modulus value estimated from the RVE model through the asymptotic expansion homogenization method and experimental data is 3.26%. Therefore, the RVE model is accurate and efficient.

Keywords 2D woven, asymptotic expansion homogenization method, CVI SiC_f/SiC composites, RVE model with porosity

1. Introduction

2D woven SiC_f/SiC composite materials have a broad application prospect in the aerospace field because of their specific strength, specific modulus, high-temperature resistance, oxidation resistance, and other excellent features. However, the complete mechanical properties of 2D woven SiC_f/SiC composite materials do not exist because of the complex manufacturing process and high cost. The mechanical properties of composite materials depend largely on the material constituents and mesoscopic structure. Therefore, the establishment of a mesoscopic model that can restore the real composite structure is the basis of accurate prediction of mechanical properties. Estimating the mechanical properties of 2D woven composites with low porosity, such as resin and metal matrix composite materials (Ref 1-5), by using the 2D representative volume element (RVE) model is an efficient method. However, during the manufacturing of 2D woven SiC_f/SiC composite materials through chemical vapor infiltration (CVI) technology, composite materials with negligible porosity cannot be obtained. The former RVE model without porosity cannot be applied to estimate the mechanical properties of 2D woven CVI SiC_f/SiC composite materials. Chateau et al. (Ref 6) discussed the influence of porosity on the elastic properties of unidirectional fiber-reinforced SiC_f/SiC composites obtained with the RVE model. The results showed an almost 50% stiffness loss on the transverse plane when porosity increased from 4 to 9%. For 2D

woven CVI SiC_f/SiC composite materials, no reasonable or effective RVE model considers the factor of porosity.

The asymptotic expansion homogenization method based on mathematical reasoning is an important method to predict the macro-equivalent mechanical properties of composites. In the 1970s, Lions (Ref 7) and Kesavan (Ref 8, 9) proposed the asymptotic expansion homogenization method to estimate the effective properties of equivalent homogenization materials. Cui et al. (Ref 10) proved that the solution of the asymptotic expansion homogenization method converges to the real solution. Hassani (Ref 11-13) introduced and comprehensively summarized the asymptotic expansion homogenization method. The prediction results obtained by the asymptotic expansion homogenization method correspond to experimental data in the application of unidirectional composites (Ref 14), 2D woven composites (Ref 15), and 3D braided composites (Ref 16).

In this study, an RVE model with porosity for 2D woven CVI SiC_f/SiC composite materials is developed according to the measurement statistics of the feature structure of scanning electron microscope (SEM) photographs and the modeling method provided by Ref 17. In accordance with CVI technology, the matrix deposition rates in each direction along the surface of fiber bundles are assumed to be the same. The basic sizes of the RVE model are determined with parameters measured and gathered from the SEM photographs. The internal porosity structure is formed naturally after the matrix model is established. The porosity of the RVE model is 17.5%, and the porosity of the composites according to the material density is 17%. The composite material is assumed to be an ideal 2D woven structure that is connected to the connection matrix between a layer and the layer structure; it can be closed by the connection matrix along the thickness direction after layer composite deposition to ensure that the composite material has a uniform distribution. The thickness of the connection matrix is determined by the porosity difference, and the final RVE model is established. The RVE model conforms to the real structure and accurately shows the location and shape of internal porosity. The mechanical properties of the 2D woven CVI SiC_f/SiC composites are estimated from the RVE model by using the verified code of the asymptotic expansion

Xiuli Shen and Longdong Gong, School of Energy and Power Engineering, Beijing University of Aeronautics and Astronautics, Beijing 100191, China. Contact e-mail: gonglongdong23@163.com.

homogenization method. The relative error of the calculated tensile modulus value and the experimental data is 3.26%. The proposed RVE model is thus accurate and efficient.

2. Tensile Test on 2D Woven SiC_f/SiC Composite Materials

A sample of the tensile test of 2D woven SiC_f/SiC composite material is shown in Fig. 1. The tensile tests were conducted in an MTS-810 electro-hydraulic servo testing machine at a strain rate of 10^{-3} /s at 20 °C and relative humidity of 50%. Figure 2 shows the tensile test sample; two extensometers (MTS 632.12C-20 and MTS 632.13C-20) are located on both sides of the MTS-810 electro-hydraulic servo testing machine. Five effective samples were identified among the six test samples.

The samples subjected to the tensile test are shown in Fig. 3. The test data are shown in Table 1. All fracture positions are in the test section, except for sample 4. Hence, the test data of sample 4 were ignored. The stress-strain plots of the samples are shown in Fig. 4. The average tensile modulus (E_1) of the 2D woven SiC_f/SiC composites is 228.8 ± 4.5 GPa.

3. RVE Model with Porosity

The SiC matrix was deposited on the woven preform, and another layer of preform was paved after deposition of the former layer was finished in the manufacturing of the 2D woven SiC_f/SiC composite material by CVI technology. In accordance with the CVI process, the following assumptions were made.

1. The 2D woven preform is uniform, and the fiber bundles with an unchanged cross-sectional shape are connected by the matrix.
2. The matrix deposition rates in each direction along the surface of the fiber bundles are the same, and they can be closed by the connection matrix along the thickness direction after layer composite deposition.

After the tensile test, the samples were divided, and their cross section was scanned with SEM to determine the inner structure of the composite material. Figure 5 shows the divided samples. The parameters of the feature structure of the RVE model were gathered and measured according to the modeling method provided by Ref 17. Figure 6 shows the real structure

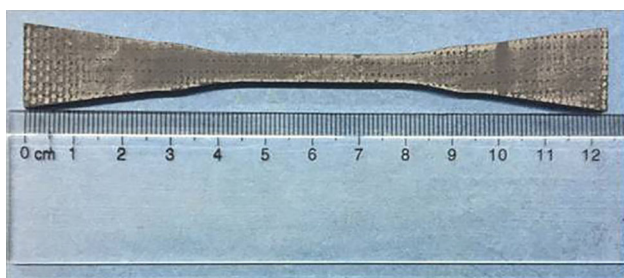


Fig. 1 Sample of the tensile test of 2D woven SiC_f/SiC composites



Fig. 2 Tensile test device



Fig. 3 Samples of the 2D woven SiC_f/SiC composites after the tensile test

of the studied 2D woven SiC_f/SiC composite material. The ideal 2D woven composite structure and the RVE model without porosity are shown in Fig. 7. Different colors are used to distinguish the fiber bundles and matrix. According to the assumptions and as shown in Fig. 6, porosity only exists in the matrix, and the matrix is dependent on the fiber bundle structure. Table 2 shows the data of the RVE model feature parameters (as shown in Fig. 8) gathered and measured from the samples of SEM photographs (as shown in Fig. 9, 10, 11).

The final feature parameters of the RVE model are shown in Table 3. The RVE model length is $l = 2 \times p$. The minimum

Table 1 Test data of tensile modulus

No.	Tensile modulus, GPa	Relative error, %
1	230	1
2	222	−3
3	222	−3
4	238	—
5	248	8
6	222	−3
Average	228.8 ± 4.5	

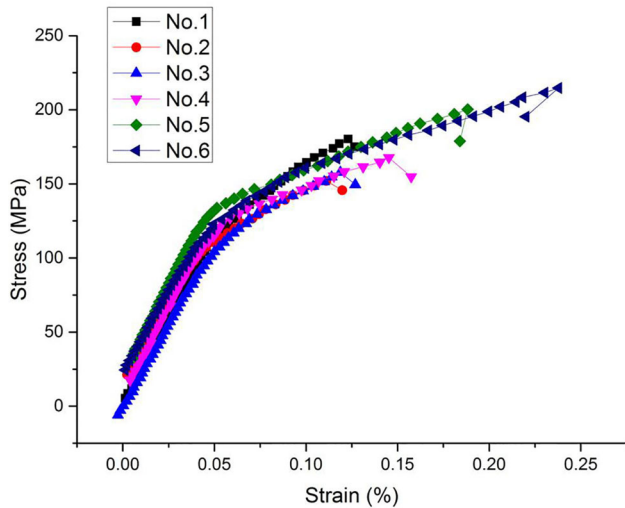


Fig. 4 Stress-strain plots of the samples

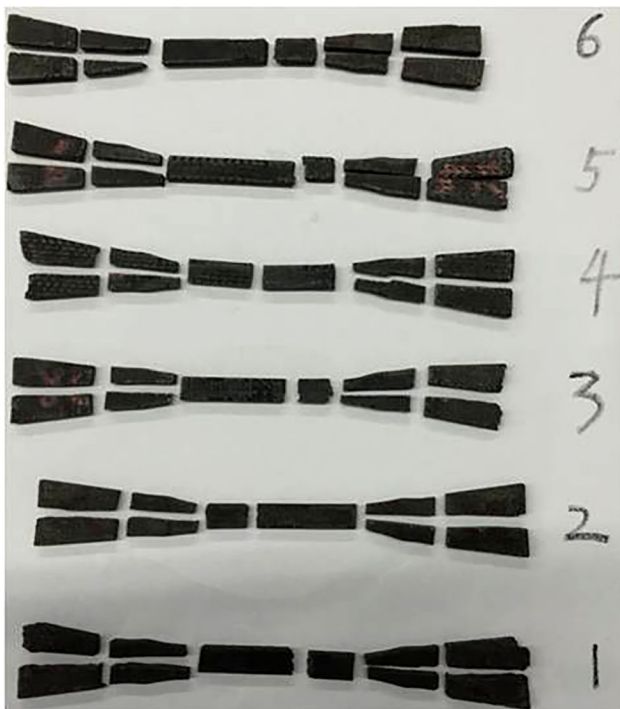


Fig. 5 Divided samples of 2D woven SiCf/SiC composites

thickness m_1 (Fig. 12) of the matrix connecting the fiber bundles is 0.02 mm, which is convenient to obtain the finite element model. The porosity of the composites calculated by

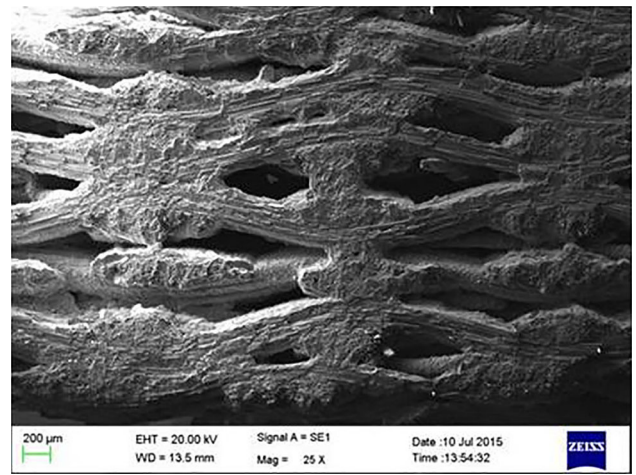


Fig. 6 Real inner structure of 2D woven SiCf/SiC composites

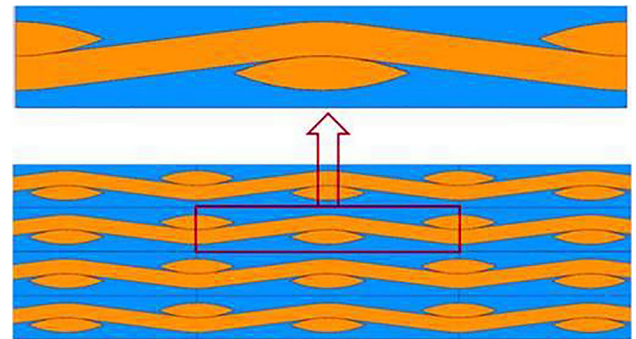


Fig. 7 Ideal 2D woven composite structure and the RVE model

the material density is 17%, whereas the porosity obtained by the model without a connection matrix between two layers is 17.5%. The connection matrix thickness m_2 (Fig. 12) is 0.012 mm, which is determined by the porosity difference. The final RVE model with porosity is shown in Fig. 12(a).

4. Asymptotic Expansion Homogenization Method and Program Verification

For periodic composites, the macro-displacement vector is a function of not only macro-coordinates x_i but also micro-coordinates y_i . Therefore, any field vector $\Phi^e(x)$ has the following form.

$$\Phi^e(x) = \Phi(x, y)$$

The ratio of the actual length of a unit vector in the micro-coordinates to the realistic length of a unit vector in the macro-coordinates is a small parameter ε ($0 < \varepsilon \ll 1$). Thus, $y = \frac{x}{\varepsilon}$. The function in the micro-coordinates is periodic because the micro-structure of composites is periodic. The field vector is

$$\Phi^e(x) = \Phi(x, y) = \Phi(x, y + Y). \quad (\text{Eq 1})$$

The derivative formula for Eq 1 is

$$\frac{\partial \Phi^e(x)}{\partial x_i} = \frac{\partial}{\partial x_i} (\Phi(x, y)) = \frac{\partial \Phi(x, y)}{\partial x_i} + \frac{1}{\varepsilon} \frac{\partial \Phi(x, y)}{\partial y_i}. \quad (\text{Eq 2})$$

Table 2 Statistics of the RVE structural parameters

No.	$w, \mu\text{m}$	$t, \mu\text{m}$	$p, \mu\text{m}$	$h, \mu\text{m}$	$s, \mu\text{m}$
1	855.53	188.39	1507.7	547.75	1232.59
2	968.65	207.51	1650	544.98	1186.08
3	1033.24	208.98	1449.77	531.7	1214.1
4	927.95	197.69	1585.85	533.24	1223.26
5	932.98	216.18	1478.21	550.04	1190
6	846.69	226.87	1562.08	526.52	1270
7	904.51	244.79	1615.11		1250
8	957.49	200.19			1093
9	916.81	215.95			1052
Average	927.09	211.84	1549.82	539.04	1190.11
Standard deviation	17.97	5.28	25.82	3.64	12.89

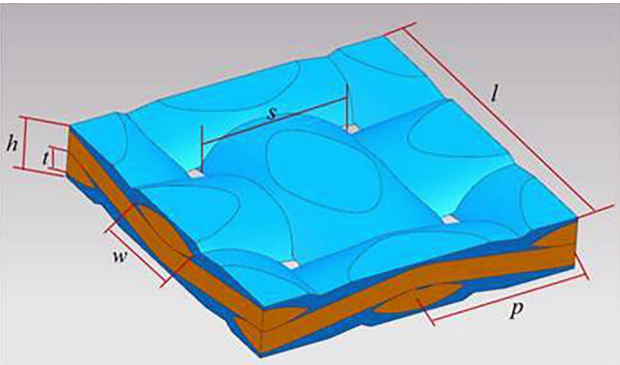


Fig. 8 Parameters of the feature structure of 2D woven composites

Through asymptotic expansion deduction, we conclude that the stiffness matrix is

$$D^H = \frac{1}{|Y|} \int_{\Omega} [D + DL_{y_i} M(y_i)] d\Omega \tag{Eq 3}$$

and

$$L_{y_i}^T D + L_{y_i}^T D L_{y_i} M(y_i) = 0, \tag{Eq 4}$$

where D^H is the macro-stiffness matrix, D is the micro-stiffness matrix, Y is the micro-period, and y_i is the micro-coordinate parameter.

The finite element form of Eq 4 is

$$Kq = F, \tag{Eq 5}$$

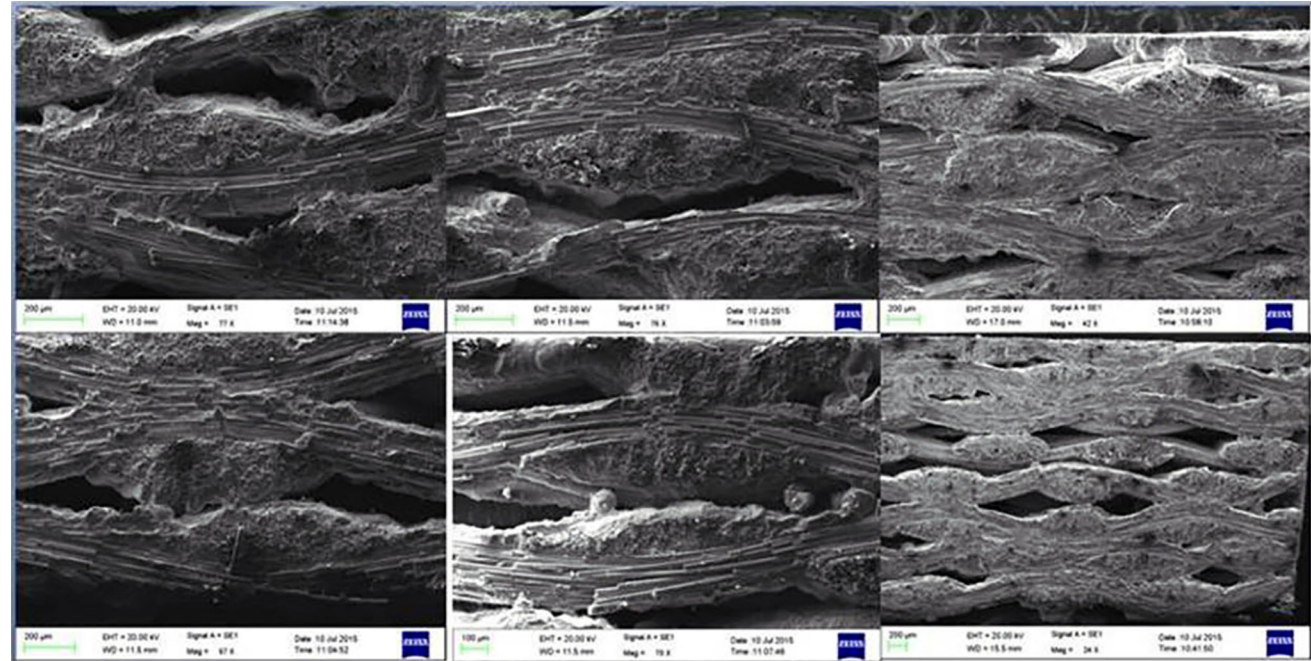


Fig. 9 Selected SEM photographs to measure the parameters of the fiber bundle cross section

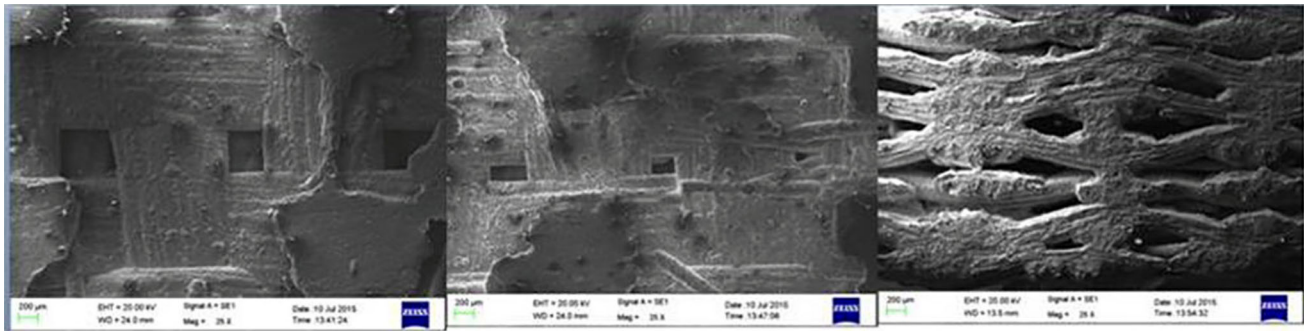


Fig. 10 Selected SEM photographs to measure the fiber distance and matrix thickness

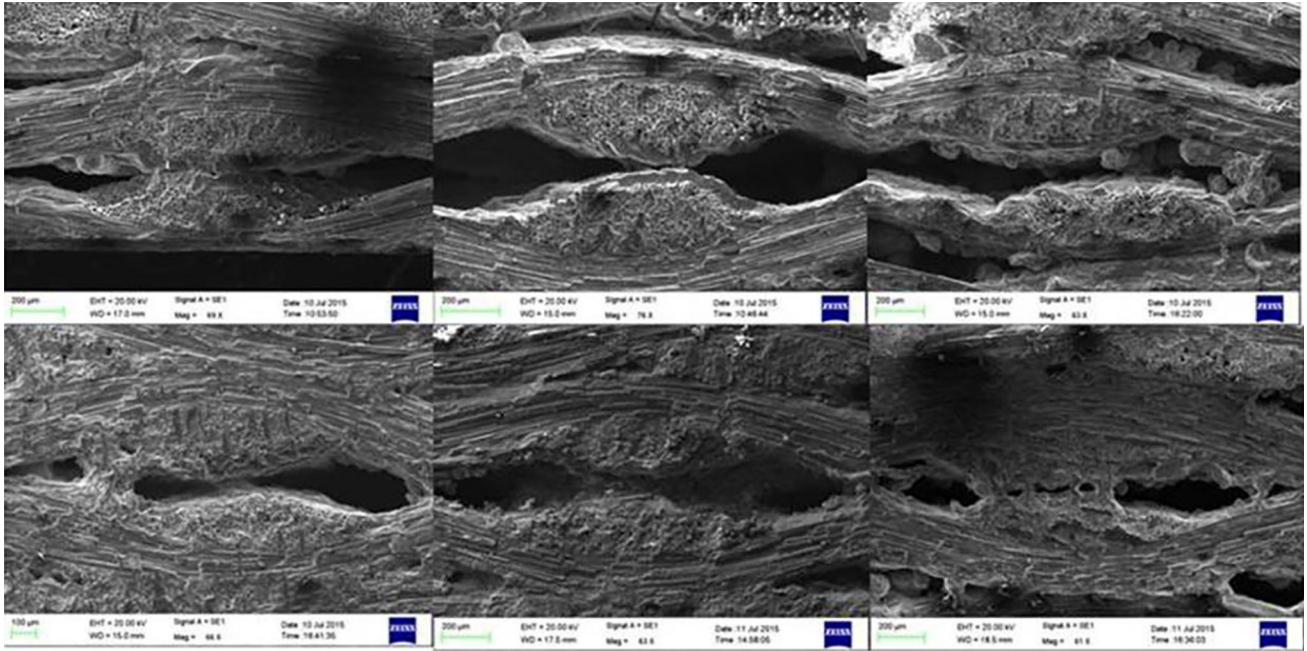


Fig. 11 Selected SEM photographs to measure the length of the RVE model

Table 3 Feature parameters of the RVE model

Feature parameters	w , mm	t , mm	p , mm	h , mm	s , mm
Value	0.93	0.21	1.55	0.54	1.19

where $K = \sum \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 B^T D B J d\zeta d\eta d\zeta$, $F = \sum \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 B^T D J d\zeta d\eta d\zeta$, q is the extended displacement vector of the nodes, extended strain matrix $B = LN$, and J is the Jacobian matrix.

$M(y)$ and q are defined as

$$M(y) = Nq. \tag{Eq 6}$$

Twenty nodes with isoparametric element shape functions were selected, and a program to estimate the macro-mechanical properties was developed with VC++6.0 software. The boron/aluminum (B/Al) composite material (Ref 18) ($E_f = 379.3\text{GPa}$, $\nu_f = 0.1$, $E_m = 68.3\text{ GPa}$, $\nu_m = 0.3$, and fiber volume of 0.47) was regarded as an example to compare with the results calculated using the finite element method (FEM) (Ref 19). The mechanical properties (E_1 is the tensile modulus along the axis

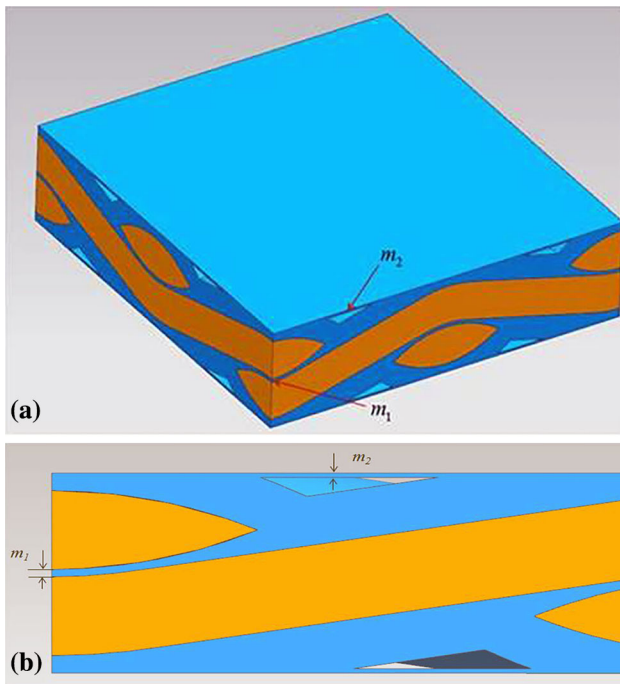


Fig. 12 RVE model with porosity for 2D woven CVI composites

Table 4 Comparison of calculated and experimental results for B/Al

	Paper	FEM (Ref 19)	Experimental data (Ref 18)
E_1 (GPa)	215.33	215	216
E_2 (GPa)	144.97	144	140
G_{12} (GPa)	54.38	57.2	52
G_{23} (GPa)	45.83	45.9	
μ_{12}	0.1946	0.19	0.29
μ_{23}	0.2551	0.29	

Table 5 Properties of the constituents of the SiC_f/SiC composites

	E, GPa	μ , GPa	ρ /g cm ³
SiC fiber bundle	200 (Ref 20)	0.25 (Ref 20)	2.7
SiC matrix	400 (Ref 20)	0.2 (Ref 20)	3.21

direction) predicted by different approaches are listed in Table 4, which shows that the developed program is correct and reliable.

5. Results and Discussion

The inner structure of the RVE model and the real structure are shown in Fig. 13. The RVE model conforms to the real structure and accurately shows the location and geometric shape of internal porosity.

Table 5 shows the properties of the constituents of the SiC_f/SiC composite material. The composites' density ρ is 2.5 g/cm³, and the fiber volume is 32%. The porosity is

$$f_v = 1 + \frac{\rho_f}{\rho_m} f_f - f_f - \frac{\rho}{\rho_m}, \quad (\text{Eq 7})$$

where ρ_m is the matrix density. According to the density, the porosity fraction is 17%. The porosity calculated from the RVE model without a connection matrix is 17.5%, which conforms well to the volume percentage obtained by density. The connection matrix was inputted to the RVE model to make the RVE model meet the periodic boundary conditions.

Table 6 shows the macro-mechanical properties (E_3 is the tensile modulus along the thickness direction) of the 2D SiC_f/SiC composite material estimated from the RVE model using the program of the asymptotic expansion homoge-

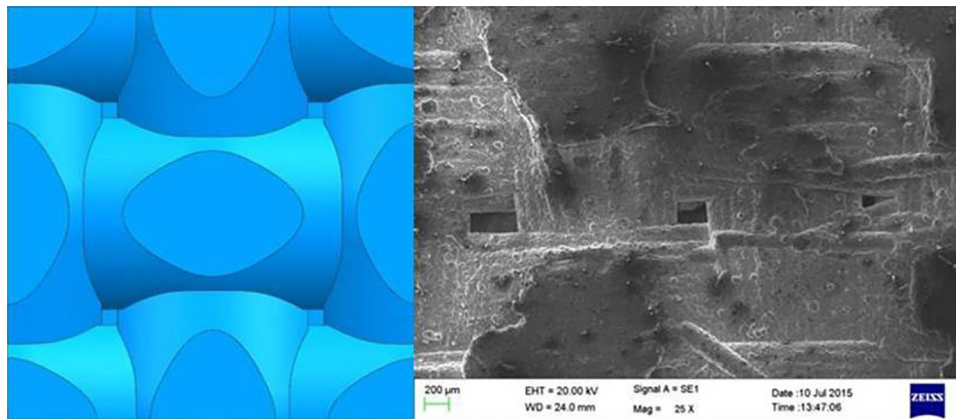


Fig. 13 Comparison of the inner structure of the RVE model and the real structure

Table 6 Comparison of the calculated and experimental results of SiC_f/SiC composites

Material parameters	E ₁ (GPa)	E ₂ (GPa)	E ₃ (GPa)	G ₁₂ (GPa)	G ₂₃ (GPa)	G ₁₃ (GPa)	μ ₁₂	μ ₁₃	μ ₂₃
Calculated value	236.3	236.3	81.40	96.92	46.37	46.34	0.20	0.21	0.21
Experimental data	228.8	...	...	...	...	...	...	...	...
Relative error	3.26%	...	...	...	...	...	...	...	...

nization method. The relative error of the calculated tensile modulus value and the experimental data is 3.26%. This result shows that the RVE model is accurate and efficient.

6. Conclusion

The RVE model with porosity presented in this paper is a reasonable RVE model for 2D woven CVI SiC_f/SiC composites and allows for the accurate prediction of the mechanical properties of these composites. The RVE model was established according to the CVI process and the measurement statistics of the feature parameters from the SEM photographs. The RVE model was verified as accurate and efficient through a comparison of the model and a real structure and a comparison of the mechanical properties obtained by the asymptotic expansion homogenization method and the tensile test.

The complete mechanical properties of 2D woven SiC_f/SiC composites, especially properties along the thickness direction, do not exist. The mechanical properties obtained from the RVE model can be utilized to analyze the components of 2D woven CVI SiC_f/SiC composite materials.

Fiber bundle was used as an isotropic material in this study. The unidirectional fiber-reinforced composite RVE model can actually be utilized to obtain the properties of real fiber bundle materials and can further improve the evaluation accuracy of mechanical properties.

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